Review plan · for the teacher

Pipelines, quality and governance: unit check

Data engineering · Level 4 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Automate the weekly report
- Checks that catch problems
- Who may see what
- Keep, archive, delete

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (20 min): Read each question aloud from the quiz sheet; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask one team to explain why it's right, using an example from the unit.
4. Wrap-up (8 min): Write the question most teams got wrong on the board, and work through it together.

Discussion

- Which report would you automate first?
- Who sees too much in our school today?
- When did we last test a restore?

Exit ticket

- One change I will propose to our head this term, and why.

Try it at school tomorrow
Pick one report your school sends, and write down its pipeline, its checks, who may see it and how long its data is kept: one line each.

Quiz sheet · for each team

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Team _____ Date _____

1 **Put a pipeline's steps in order.**

- [ ] Load
- [ ] Extract
- [ ] Log the run
- [ ] Transform

2 **A load fails halfway and the pipeline reruns. Which load gives the right report?**

◯ One that adds the rows again ◯ One that replaces the day's rows

◯ Either: the computer sorts it out

3 **Retrying fixes a register that says "?" instead of P or A.**

◯ True ◯ False

4 **Match each check to the SQL that runs it.**

In range	• • • •	WHERE mm IS NULL
Not empty	• • • •	GROUP BY station, day HAVING COUNT(*) > 1
No repeats	• • • •	WHERE mm < 0 OR mm > 300

5 **This check never finds the missing marks. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT pupil_id
2  FROM marks
3  WHERE mark = NULL;
```

6 **Every check passes on today's batch. What can you truthfully say?**

◯ No known problem was found ◯ The checks need switching off ◯ The data is correct

7 **Which are safe ways to share this term's results with the district? Choose all that apply.**

- [ ] School-level results by subject
- [ ] Averages per class, hiding groups under five
- [ ] A class of three's average next to its special needs
- [ ] Every pupil's mark with numbers instead of names

8 **Fill the gaps: averages per class, without any class of fewer than 5 pupils.**

```
SELECT class, ROUND(AVG(mark), 1) FROM marks
GROUP BY class
 _________ COUNT(*) _________ 5;
```

Word bank: `>=` `<` `HAVING` `WHERE`

9 **Complete the backup rule.**

Keep _____ copies, on _____ kinds of storage, with _____ away from the school, and test a restore every term.

Word bank: four · three · five · one · two

10 **The school is giving away five old laptops. What should happen first?**

◯ Wipe each one with a full reset, and log it ◯ Delete the pupils' folders

◯ Nothing: they're too old to matter

Answers · for the teacher

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1 **Top to bottom: 3, 1, 4, 2**

Extract reads the data, transform cleans and counts it, load writes the result, and the log records what happened.

2 **One that replaces the day's rows**

Replacing the day's rows is idempotent: however many reruns, the report is the same.

3 **False**

Bad data fails the same way on every retry.

4 **In range — WHERE mm < 0 OR mm > 300**
Not empty — WHERE mm IS NULL
No repeats — GROUP BY station, day HAVING COUNT(*) > 1

Each check is a query that returns the bad rows: out-of-range amounts, empty values, and station-days with more than one reading.

5 **Line 3:** `WHERE mark IS NULL;`

Nothing equals NULL, so = NULL is never true.

6 **No known problem was found**

Checks catch only the problems someone thought of.

7 **Averages per class, hiding groups under five; School-level results by subject**

Summaries with the small-group rule answer the district without describing any child.

8

```
SELECT class, ROUND(AVG(mark), 1) FROM marks
GROUP BY class
HAVING COUNT(*) >= 5;
```

HAVING filters the groups after they're counted, and >= 5 keeps only classes of five or more.

9 **Keep three copies, on two kinds of storage, with one away from the school, and test a restore every term.**

3-2-1: three copies, two kinds of storage, one away from the school, and a restore test, because an untested backup is only a hope.

10 **Wipe each one with a full reset, and log it**

Deleted files can often be recovered. A full reset that also cleans the drive (or destroying the drive), written in the log, is deleting on purpose.